

Names as Data

The field every record about a person carries, and the three jobs it is asked to do

84–355 Democracy’s Data

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Almost every record in this book carries a name. The voter file has one, the campaign filing has one, the court docket, the jury questionnaire, the police stop, the lobbying registration. Some of these records carry nothing else about the person: no age, no address, no race, no sex.

So the name gets asked to do work. Three jobs, over and over:

- 1. **Identify.** Two records carry the same name, so they are the same person.
- 2. **Stand in for sex**, when the file does not record it.
- 3. **Stand in for race**, when the file does not record it — which is most of the time, because most American records of a person do not.

None of these is a job a name can do properly, and the interesting question is not whether they work but *how badly, and in which direction*. In April 2026 the Census Bureau published the file that lets you check. It is two tabulations of every American name, from the 2020 census: one of first names and one of last names. This chapter is about what is in that file, and what had to be done to a name before it could be counted. And about what each of the three jobs above looks like when you measure it instead of assuming it.

01 What arrives

Frequently Occurring First Names in the 2020 Census by Sex						
FIRST NAME	RANK	FREQUENCY (COUNT)	PROPORTION PER 100,000 POPULATION	CUMULATIVE PROPORTION	MALE	FEMALE
MICHAEL	1	3476721	1151.1119	1151.1119	3469434	777287
JOHN	2	3125123	1034.7009	2185.8128	3118620	606503
JAMES	3	2996421	992.0888	3177.9016	2992500	393921
DAVID	4	2813017	931.3653	4109.2669	2810463	492554
ROBERT	5	2759339	913.5930	5022.8599	2756604	292735
... 53,610 rows omitted ...						
ALL OTHER NAMES		18794706	6222.7628	100000.0000	7768920	11025786

That is the first-name-by-sex table, opened. Two title rows above the column names, one row per name, and the proportion columns cut here from seventeen digits to four. A second file gives the same names by race and Hispanic origin; a third and fourth give the last names.

Table	Names published	People covered	Share of the 2020 census	Swept into ALL OTHER NAMES
First names	53,615	302,031,536	91.1%	18,794,706
Last names	156,621	298,898,231	90.2%	36,257,637

Three things in that table are worth stopping on before anything is computed from it.

The threshold. A name is listed if about a hundred people had it. Everyone else, 18,794,706 people by first name, is swept into a single final row called ALL OTHER NAMES. In the exhibit above that row has an empty rank cell, so a reader who filters out rows without a rank deletes 18.8 million people and is told nothing.

The coverage. A first name was recorded for 91.1% of the people the 2020 census enumerated. The other 29,417,745 have no usable first name in this file at all, and they are not a random draw of the country. They are concentrated wherever the census had to fall back on administrative records or proxy answers to count somebody, which is where coverage is worst to begin with.

The counts have been noised. These are 2020 census tabulations, so they went through the same disclosure-avoidance machinery as everything else from that census. What that did here is stranger than usual, and it gets its own section at the end.

02 Before a name could be counted, it had to be made

The Bureau's own methodology brief is the most useful thing on the page it was published from, and this is the part of it nobody quotes.

The unedited 2020 returns contained about **6.0 million distinct first names**. Editing removed roughly 800,000 of them and revised millions more, leaving 5.2 million. The largest single problem was not misspelling. It was that over **ten million first-name entries were not names**: PERSON, CHILD, DAUGHTER, GRANDSON, ELDERLY, RESIDENT, TENANT, HOUSEHOLDER, MRS SMITH, EIGHTEEN, TWENTY-TWO, CHILD 2, PERSONA DOS. Left alone, PERSON would have been one of the most common first names in the United States.

Two details from that clean-up are worth carrying into every other chapter.

Punctuation and structure were removed, not preserved. All punctuation, numbers and spaces go, and so do suffixes: JR, SR, III, THIRD. MICHAEL, MI CH AEL, MICHAELJR and MICHAEL III are all counted as MICHAEL. That is the right decision for counting names, and it is exactly the decision a matching program has to make too. A matching program usually does not make it, and so treats those four as four different people.

Ñ did not survive the trip. In the 2020 returns, Spanish names containing Ñ arrived with the character turned into a question mark, so PEÑA came through as PE?A. The tables carry only the 26 English letters, so the Bureau corrected the question mark to N, and PEÑA is published as PENA. The name was damaged upstream, in transcription, and the repair is a second decision layered on the damage. Neither is hidden; both are irreversible.

And the clean-up is not finished. The Bureau states that the edits caught more than 95 percent of nonname entries. So some are still in the table. Real names that are also ordinary words, like BABY, CHILD, PERSON, SON and YOUNG, may be inflated, and the frequency published for DOE "is surely too high."

Before you read on, commit to a number. There are 156,621 last names in this file. **How many of them would you need — the most common ones — to cover half of everybody in it?** Write the number down. Then do the same for first names, of which there are 53,615.

03 A name is a distribution, not a label

Rank	First name	People	Last name	People
1	MICHAEL	3,476,721	SMITH	2,369,644
2	JOHN	3,125,123	JOHNSON	1,858,234
3	JAMES	2,996,421	WILLIAMS	1,561,395
4	DAVID	2,813,017	BROWN	1,386,083
5	ROBERT	2,759,339	JONES	1,383,250
6	WILLIAM	2,240,114	GARCIA	1,149,510
7	MARY	1,768,343	MILLER	1,129,186
8	MARIA	1,652,966	RODRIGUEZ	1,085,838
9	DANIEL	1,607,193	DAVIS	1,074,448
10	JOSEPH	1,600,667	MARTINEZ	1,039,848

The answers to the question above are **317 first names** and **2,412 last names**. Out of 53,615 and 156,621 published, and out of the 5.2 and 7.8 million the census actually collected.

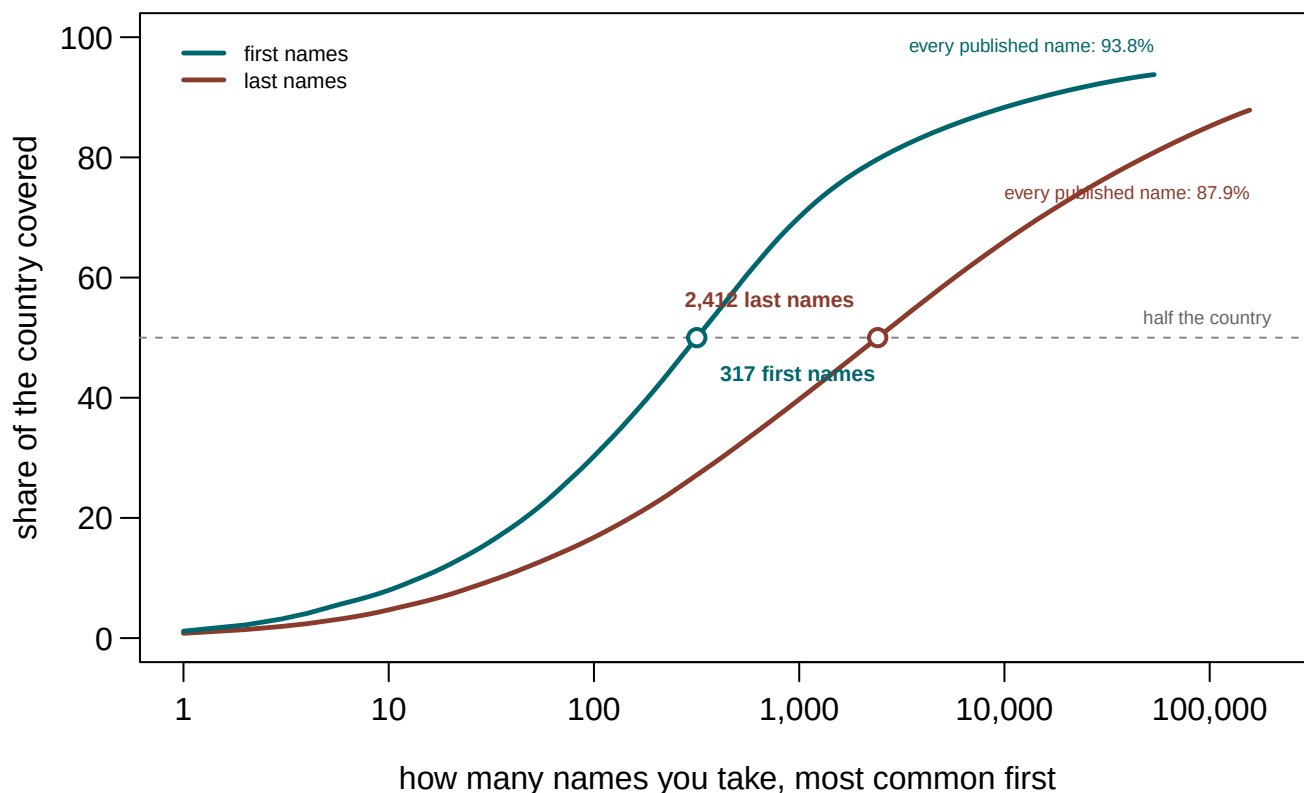


Figure 1. How much of the country the most common names cover. In the browser version, moving the pointer across the figure reads a rank off it.

Neither curve reaches the top, and that is the residual again, the share left over, rather than a drawing error. Take **every** published first name, all 53,615 of them, and you have covered 93.8% of the people in the file. Every published surname covers 87.9%. The rest hold names that fewer than a hundred Americans share, and the file will not tell you what they are.

The most common	Of first names	Of last names
10 names	8.0%	4.7%
100 names	30.3%	16.8%
1,000 names	70.0%	39.7%
10,000 names	88.3%	66.0%

First names are the more concentrated of the two, and by a lot: a thousand of them cover 70.0% of Americans, where a thousand surnames cover 39.7%. That ordering runs against the intuition that first names are where individuality lives. It has a plain cause. A surname is inherited from a limited stock, and American surnames come from every immigration history the country has had. A first name is chosen, and chosen from fashion, which is a much smaller menu than the world's surname stock.

Two summaries of the same distribution, and only one of them survives

There is a standard way to say how many names there “really” are. Both of these have a name in the ecology literature and both get used on distributions like this one, and this file separates them cleanly.

The first is the chance that **two people picked at random share a name** — Simpson’s index, the sum of the squared shares. That number is 1 in 634 for first names and 1 in 1,911 for last names.

The second is the **effective number of names**: how many equally common names would be as hard to guess as the real distribution. It is the exponential of the entropy, and it is the number people reach for when they want to say “53,615, but really only about N .”

Only the first one can be computed from this file, and the reason is the residual line. Everything below the threshold is one lump: millions of names the file does not list.

For the collision probability that does not matter. Each of those names is held by fewer than a hundred people, so a bound falls straight out. The whole unlisted tail can account for at most 0.001% of the first-name figure, and 0.008% of the last-name one. For the effective number it matters entirely. That figure is a statement about how the tail is spread, and the tail is what is missing. All this file supports is a bracket:

Names	Published	Effective number	Chance two people share one
First names	53,615	between 3,780 and 5,034	1 in 634
Last names	156,621	between 27,810 and 48,619	1 in 1,911

The brackets are the two extremes the tail could take: every unlisted name held by the maximum hundred people, or every one held by a single person. The truth is somewhere between, and this file cannot say where. That is not a defect in the file. It is a demonstration that **the choice of summary statistic decides whether a truncated table can answer your question** — and the one that survives is the one the next section needs.

04 The first job: identifying a person

Two random Americans	Chance
two people share a first name	1 in 634
two people share a last name	1 in 1,911
two people share both, if the two are unrelated	1 in 1,212,277
two people share both, allowing correlation through race	1 in 1,082,518

A first name and a last name together are worth about **one in 1,212,277**. That is the number a matching program is quietly relying on. It is arrived at by multiplying the two separate figures together, which is only legitimate if a person’s first name tells you nothing about their last name.

It obviously does. These files publish no joint table of first name against last name, so the size of that error cannot be measured here. A floor on it can be: let the two names be correlated through the six census race categories and through nothing else, and shared full names get 12% more likely, at 1 in 1,082,518. Six categories is a coarse instrument for this. The real correlation is finer than any category the file publishes. It runs through national origin, when a family arrived, and generation, and it is the reason NGUYEN and GARCIA do not draw their first names from the same pot.

One thing to keep straight about all four of those numbers: each is the chance for **one pair of people**. It is not the chance of finding a match in a list, which depends on how long the list is and is a different calculation entirely — the one the false matches chapter does.

The two fields are not even distinct

Name	As a first name	As a last name
RUSSELL	202,735	211,729
KIM	246,384	274,068
GARRETT	116,551	102,634
MORGAN	235,724	276,366
DEAN	132,116	109,762
RAY	107,593	130,289
SPENCER	106,323	135,510
GRANT	109,200	139,862
HARPER	92,376	119,191
ROSE	196,309	149,763
OLIVER	146,493	109,817
COLE	134,580	187,989

14,926 names appear in both tables. **75.1% of Americans have a first name that is also somebody's surname**, and 50.0% have a surname that is also somebody's first name. The twelve above are the most-used names that are split most evenly between the two roles.

This is the reason a swapped-column bug in a name file is so hard to see. Suppose `first` and `last` are swapped for part of a load. Three-quarters of the affected rows still hold a perfectly ordinary name for the column it has landed in, and no check that asks whether a field “looks like a name” will fire.

05 The second job: standing in for sex

The first-name table is the only place the Bureau attaches sex to a name. It answers the question analysts actually ask: assign every person the majority sex of their first name, and how many do you get wrong?

97.51% right. Which is **7,042,013 people wrong** — a population between Washington and Massachusetts, misfiled by a rule that is correct about 39 times out of 40.

Names where the minority sex is at least	Names	People	Share of the country
1%	30,509	62,363,914	22.0%
5%	16,546	30,686,800	10.8%
10%	11,895	21,191,006	7.5%
25%	5,972	8,212,126	2.9%
40%	2,222	2,882,408	1.0%

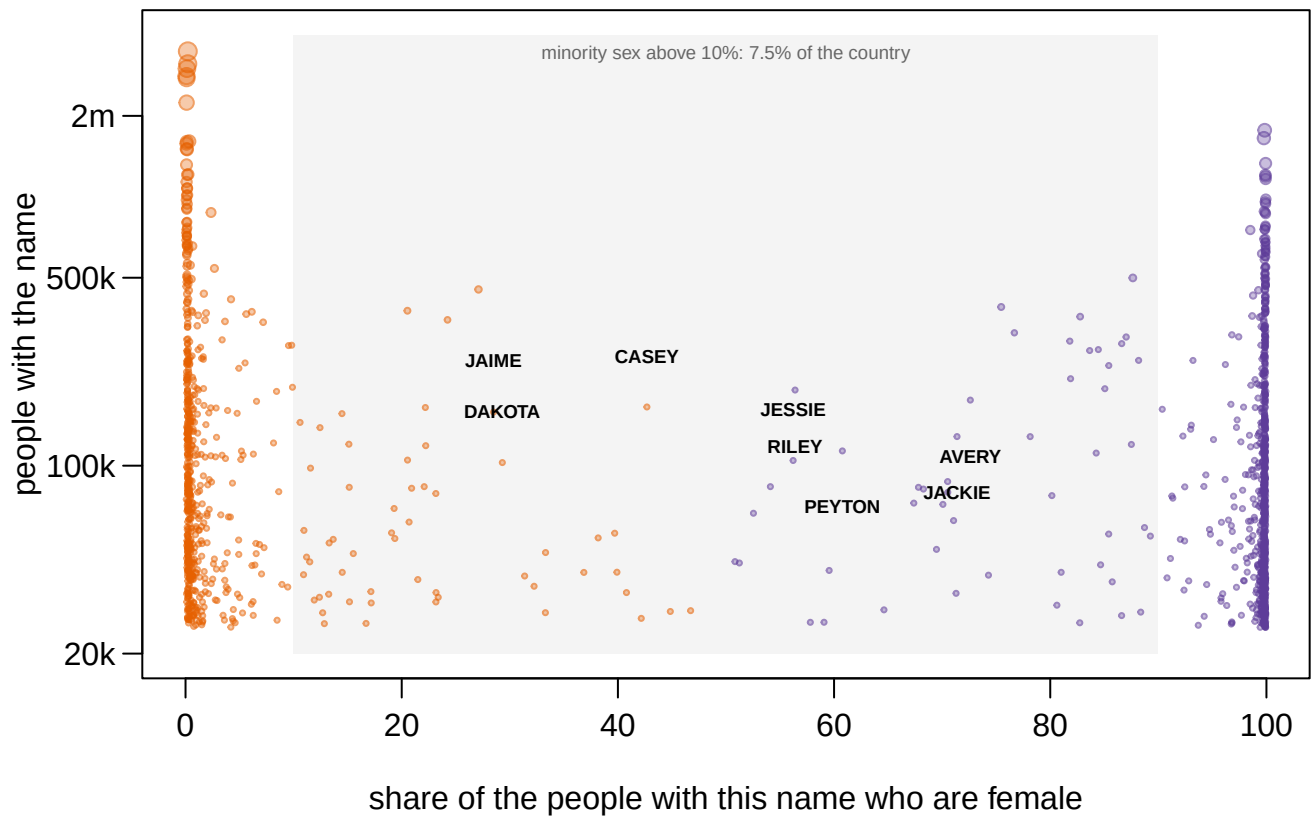


Figure 2. Every first name held by 25,000 people or more – 1,394 names, a declared cut, below which the figure is a solid cloud. Point area is proportional to the number of people. The browser version names any point you hover.

Name	People	Female	Male
JESSIE	104,536	56.2%	43.8%
RILEY	191,103	56.4%	43.6%
CASEY	165,300	42.7%	57.3%
PEYTON	113,474	60.8%	39.2%
DAKOTA	102,684	29.3%	70.7%
JACKIE	128,298	71.4%	28.6%
JAIME	158,232	28.5%	71.5%
AVERY	175,333	72.6%	27.4%
JORDAN	452,330	27.1%	72.9%

Name	People	Female	Male
TAYLOR	389,216	75.5%	24.5%
ANGEL	348,575	24.2%	75.8%
JAMIE	312,010	76.7%	23.3%

These twelve are the most-used first names, 102,684 bearers and up, that come closest to an even split. JORDAN is 27.1% female and TAYLOR is 75.5%, close to mirror images of each other. A rule that assigns each of them to its majority is making the same size of mistake in opposite directions. Between them the two names account for 841,546 people.

Two limits on all of this, and both are about the question rather than the arithmetic. The census item is **sex**, asked as two boxes. It is not gender, it does not allow a third answer, and nothing in this file speaks to people for whom the two differ.

And the table is a snapshot of everyone alive in 2020. A name whose usage shifted, and the names in that list are mostly names that shifted, is being described by an average taken across the shift.

06 The third job: standing in for race

Here the two kinds of name behave differently, and the difference is the reason this book's race-inference chapters use surnames.

Put both on one scale by asking how much a name tells you, in bits. Before you hear a person's name, your uncertainty about which of the six census categories they fall into is some number of bits. After you hear it, it is smaller. The difference is what the name carried.

Given only a	Modal guess is right	Bits before	Bits after	Uncertainty removed
First name	74.2%	1.64	1.19	27.7%
Last name	79.3%	1.70	0.94	44.6%

A last name removes 44.6% of the uncertainty about a person's race; a first name removes 27.7%. That is the whole justification for surname-based inference in one line, and it is measured rather than assumed. It is also not nothing for first names, since the résumé-audit literature is built on the fact that a first name carries a signal. But if you are going to pick one field, the surname is the one that carries more.

Now the ceiling, which is the part that sets up the chapter after this one.

Group	Most distinctive first name	Most distinctive last name	Group share of the file
white	BETH (95.0%)	YODER (96.7%)	50.4%
Black	TYRONE (81.7%)	WASHINGTON (83.6%)	16.8%
Am. Indian	CHEYENNE (3.2%)	HUNT (3.7%)	0.7%

Group	Most distinctive first name	Most distinctive last name	Group share of the file
Asian/PI	MOHAMMAD (56.9%)	VANG (96.4%)	5.0%
two or more	KAI (19.5%)	WASHINGTON (6.0%)	3.4%
Hispanic	GUADALUPE (98.5%)	VELAZQUEZ (95.2%)	23.7%

The most distinctive surname for white Americans is 96.7% white. The most distinctive one for American Indians is 3.7% American Indian. Both are the strongest signal the file contains for their group, and they are two orders of magnitude apart in what they can support.

Nothing is wrong with the names. A name cannot outrun the base rate: if a group is 0.7% of the file, then even a name whose bearers are enriched fifty-fold for that group is still mostly not that group. So a name alone will never identify the members of a small group, however distinctive the name is. Any method that tries has to bring something else to the problem, most often where the person lives. That is what surname-based inference does with this file.

One warning the Bureau prints and this section has to repeat. These tables are not a source for the racial composition of the United States. The “bits before” figures above are the composition of *the people in this file*, which is the right baseline for measuring what a name adds and the wrong number to quote as a national statistic.

07 Look a name up

First name	Rank	People	Female	Largest group	Also a surname
MICHAEL	1	3,476,715	0.2%	white 79.0%	yes
MARIA	8	1,652,964	99.8%	Hispanic 81.3%	yes
JORDAN	101	452,333	27.1%	white 61.0%	yes
TAYLOR	119	389,227	75.5%	white 77.0%	yes

Figure 3. In the browser, a box you can type any name into. It gives the rank, how many people hold it, the sex split if it is a first name, the racial composition, and whether it is also used the other way round. The cut is declared: the 2,000 most common names of each kind, which is 78.1% of Americans by first name and 47.8% by surname. Print gets four rows of it instead.

08 The same names, counted at birth instead

The census file cannot tell you when a name was given, and that is usually the first thing worth knowing about one. DOROTHY and AIDEN are not two kinds of person. They are two decades. A snapshot of everyone alive in 2020 holds every decade at once and separates none of them.

A second file does nothing else. The Social Security Administration publishes the name on every application for a card, arranged by the year the applicant was born, back to 1880. It is the only source in this chapter that is not the census, and it is here because the two files count different things.

That difference decides which questions each one can answer, so it is worth a pair of words. A **stock** is how many there are right now. A **flow** is how many arrived this year. The census hands you a stock of people, Social Security hands you a flow of births, and the two behave nothing alike when something happens to a name.

Where this second file comes from is worth pausing on, because nobody meant to build it. **Nobody was studying names.** The Social Security Administration writes down a name and a year of birth because it has to know who is who and when they qualify. The birth file is a byproduct of issuing Social Security cards, tabulated and published by the agency's actuaries, whose business is birth cohorts rather than baby names. No question was ever asked of anyone. The file exists because an administrative machine cannot run without recording names, and it covers exactly what that machine happened to touch.

What it costs to get. Nothing, with one flourish. The file sits on the agency's public site, no key and no account, but the server refuses a plain scripted request and answers one that claims to be a browser. A person can always have it; a program has to dress as a person.

None of that machinery was pointed at fashion. It records fashion anyway, faster than any census can, because a flow shows a shock in the year it lands.

Something happened to ALEXA. Amazon announced the Echo in November 2014 and put it on general sale in 2015, and the name has been falling away from babies ever since. In 2025 it went to 243 girls in the whole country, ranked 1,029, down 96% from where it stood ten years earlier.

The part usually left out is what happened first. 2015 is the best year the name has ever had: rank 32 among girls, its highest in 146 years of records, with 6,053 girls given it. The launch year was the peak. Parents appear to have heard the name, liked it, and then watched it become the thing you say to a speaker in the kitchen.

Before you read on, commit to a number. The Bureau published a first-name table from the 2020 census. Suppose it had published one from 2010 as well. **Which way would ALEXA have moved between the two, and by how much?** Write it down before you look at the next figure.

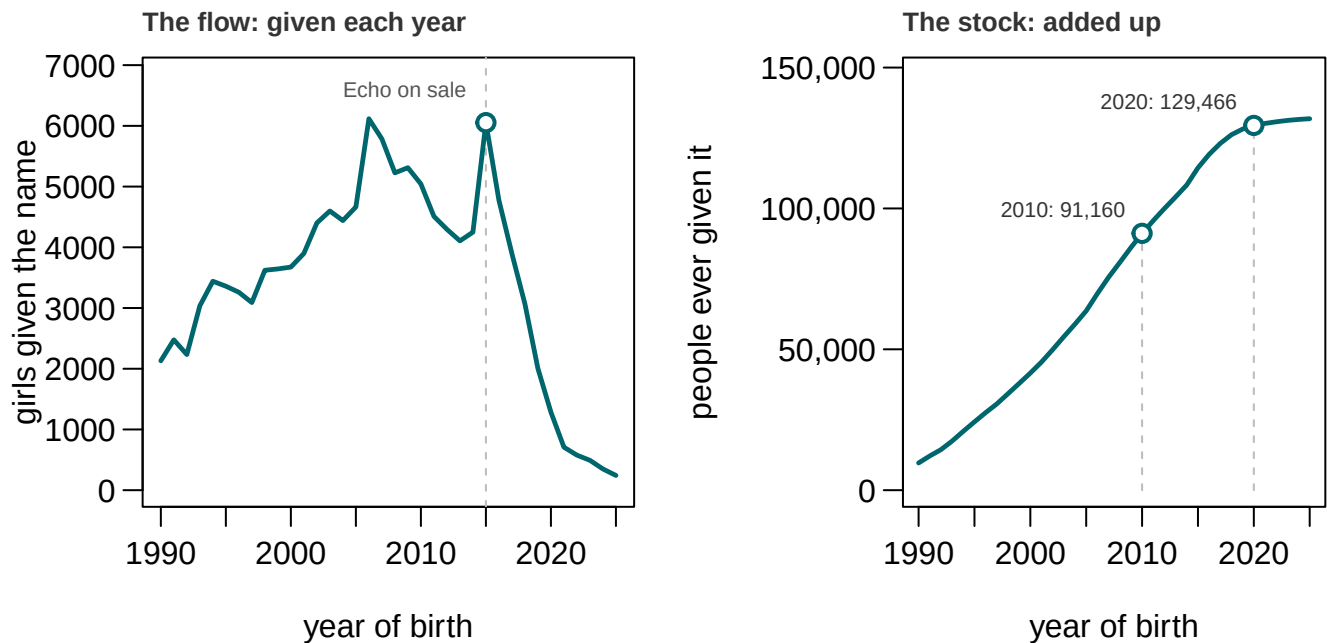


Figure 4. ALEXA, both sexes, from the Social Security birth file. The left panel is how many were given the name each year. The right panel is the same numbers added up, which is what a census counts. The left panel falls 96% and the right panel does not fall at all.

The census would have shown the name **rising by 42%**.

Nothing about that is a trick. Everyone given the name before the collapse was alive in 2020 and got counted in 2020. The babies born during the collapse were added on top of them, 38,306 more between the two censuses. A fall in the flow does not empty the stock. It only slows what is being added.

That is the rule to carry out of this section, and it is not about names. **A stock absorbs a shock and a flow records it.** Houses, registered voters, guns, people: anything counted as a stock will show you a change years late and smaller than it really was. Leaving the stock of a name takes a death, which for a name given to babies is most of a lifetime away.

A flow moves at the other speed, and ALEXA is not a special case. Three more names had something happen to them, and the birth file caught every one within about a year of the event.

Name	What happened	Year	Given the year before	Five years later	Change
KATRINA	the hurricane made landfall	2005	1,174	314	-73%
SIRI	Apple shipped Siri with the iPhone 4S	2011	111	24	-78%
ISIS	the group took Mosul	2014	501	68	-86%
ALEXA	Amazon put the Echo on general sale	2015	4,249	1,285	-70%

Read that table carefully, because it invites more than it can carry. It shows four names falling after four events. It does not show that the events caused the falls. Names go out of fashion on their own, constantly, and the table has no column for what these four would have done anyway. ISIS is the useful warning: it bottomed out at 52 girls in 2017 and was back to 230 by 2024, which is not what the permanent ruin of a name looks like.

There is also a plainer reason you cannot run that comparison yourself. **The Bureau has never published a first-name table for 2010.** The 2010 release is surnames only, and first names arrived for the first time with the 2020 one, so there is no earlier vintage to hold this against. The 2010 to 2020 comparison that does exist is the surname one, and the surnames chapter runs it.

A threshold is in the way as well, and it points the same direction. To be published in 2020 a first name had to occur at least a hundred times in **both** 2010 and 2020. A name that arrived during the 2010s is missing from the file by design, which means the census tables are least able to see exactly the names that were doing something interesting.

Which names are arriving

The fastest-rising names in America are old names. These ten gained the most ground between 2013–2015 and 2023–2025, measured as a share of each year's births. The last two columns are the year each was previously at its busiest, and how many babies got it that year.

Name	Given a year then	Given a year now	Busiest earlier year	Given that year
THEODORE	3,271	12,183	1919	3,441
MATEO	4,127	11,241	1975	42
OLIVER	9,481	15,068	1918	1,258
LUCA	2,345	8,258	1976	26
EZRA	3,679	8,844	1921	170
LUNA	2,302	7,040	1921	61
MAVERICK	1,839	6,584	1958	88
LIAM	18,375	21,356	1979	118
AMELIA	8,917	12,624	1917	1,160
THIAGO	797	5,223	—	—

THEODORE is the clearest case. It was last this common in 1919, and almost everyone who received it then is gone. A name can be new to a parent and a century old in the file.

All 10 of them are in the census table, and the threshold from a moment ago is why. A revival clears it without trouble, because the earlier cohort was alive in 2010 and got counted. What cannot clear it is a name that did not exist yet.

Name	First year in the file	People born by 2010	Babies since 2023
KEHLANI	2015	0	5,834
OAKLYNN	2008	5	5,731
WRENLEY	2013	0	5,549
CATALEYA	2011	0	3,693
KYLO	1979	11	2,493
WRENLEE	2016	0	2,242
ELOWYN	2008	13	2,146

Name	First year in the file	People born by 2010	Babies since 2023
KYLIAN	2002	51	2,079
KAIZEN	2004	81	2,009
HAISLEY	2002	49	1,958

86 names were given to 500 or more babies in the last three years and have no row in the census file at all. That is 100,993 children whose first name the 2020 tables cannot report, led by KEHLANI. Several of the ten are recognisable from film and television, and the years in the second column will line up with whatever you remember. Resist finishing that thought: the shock table above is the reason.

This is the same gap as ALEXA, running the other way. ALEXA is in the census file because it was already a common name when the assistant took the name over. The names arriving now are not, and will not be until a census counts the children who have them. A file that reports the stock is always reporting the last generation's fashions.

Holding the two files against each other

The two files agree about as well as two files counting different things can. 129,466 people have been given the name ALEXA since records begin. The 2020 census counts 115,935 of them, which is 90%. The missing tenth is deaths, people who have left the country, people who go by something else, and the 9% of the census with no usable first name at all.

Name	Median birth year	Ever given the name	Counted alive in 2020	Share still counted
ALEXA	2006	129,466	115,935	90%
EMMA	2004	708,166	498,221	70%
TAYLOR	1997	435,743	389,216	89%
JENNIFER	1978	1,473,629	1,331,077	90%
DOROTHY	1927	1,112,489	304,932	27%

The last row is a warning about the other four. That 90% works for ALEXA because it is a young name and almost nobody who has it has died yet. DOROTHY was given mostly in the 1920s, and only 27% of the people who received it are still here to be counted. Comparing the two files tells you about mortality as much as about names, and which one you are looking at depends entirely on the name.

One name in the lookup above has no birth record at all. The census gives REF to 14,848 people, and Social Security has never once registered five babies with it in a year. Either those people were all born outside the United States, or REF is not a name — and after PERSON, CHILD and TENANT, the second is the way to bet.

09 Look your own name up in the other file

Name	Median birth year	Ever given the name	Counted alive in 2020	Census rank
MARY	1938	4,156,654	1,768,343	7
DAVID	1963	3,689,613	2,813,017	4
JOSHUA	1991	1,253,882	939,718	30
LIAM	2017	359,085	218,131	257

Figure 5. In the browser, a box you can type your own name into: how often it was given in each year of birth since 1880, against how many people the 2020 census counts with it now. The line is a share of that year's births rather than a count, which is what makes the shaded early years readable at all. A declared cut — the 1,000 most common first names in the census plus the risers and the newcomers above, 1,090 names in all, which is 70.1% of Americans. The 86 newcomers add nothing to that percentage, because the census cannot see them. Print gets four rows of it instead.

Start with your own name, and find the year it peaked. For most readers that will be close to the year they were born. The buttons under the box are there for the three shapes this section has been describing, and they are worth clicking in order.

THEODORE and the two risers beside it climb twice, a century apart, and the second climb has already gone past the first. Its best year on record is 2025, which is this year. MARY and CHARLES do the opposite. They were at their most common in 1891 and 1880, longer ago than anyone now alive, and neither has recovered since. ALEXA does neither, because it is the one name here that was interrupted.

KEHLANI is the fourth shape and the one with something missing. Its curve starts at the floor and goes almost straight up, and where the card should report a census count it reports that there is no row to read. That is not a gap in the figure. It is the threshold from two pages ago, seen from the side of the file that loses by it, and 86 of the names in the box are in the same position.

Then compare the two numbers on the card for MARY or CHARLES. The census rank is a stock, and it is nothing like the rank the name held in the years it was actually being given.

10 The same census, counted twice, disagreeing with itself

One last thing about the file, and it is the kind of thing that is invisible until you hold two of its tables side by side.

The Bureau published first names by sex and first names by race as separate products, ran the disclosure-avoidance noise **independently** into each, and then derived each table's total from its own noisy cells. The consequence:

Rank	Name	In the sex table	In the race table	Gap
1	MICHAEL	3,476,721	3,476,715	6
2	JOHN	3,125,123	3,125,114	9
3	JAMES	2,996,421	2,996,420	1
4	DAVID	2,813,017	2,813,019	-2
5	ROBERT	2,759,339	2,759,332	7
6	WILLIAM	2,240,114	2,240,112	2
7	MARY	1,768,343	1,768,337	6
8	MARIA	1,652,966	1,652,964	2
9	DANIEL	1,607,193	1,607,193	0
10	JOSEPH	1,600,667	1,600,664	3

49,089 of 53,615 first names, 91.6% of them, have a different total in the two files. The two tables disagree by 20 people at the widest (MELVY), and they even disagree about the residual line, by 3.

Gap between the two tables	First names
0 people	4,526
1 people	8,849
2 people	8,469
3 to 5 people	19,026
6 to 10 people	11,468
more than 10 people	1,277

Neither file is wrong. There is no fact of the matter about how many Americans are named MICHAEL that survives publication, because publication is what added the noise. A question asked twice of a noisy process gets two answers. The Bureau documents this in the methodology brief for each product. Almost nobody who downloads one file reads the brief for the other.

The same mechanism has two more visible effects. The threshold for publication is a hundred people. And yet 155 first names and 617 last names are published with totals below it, the smallest at 91, because the noise carried them under after the cut was made. And the noise pushed some small cells below zero:

Table	Rows with a negative cell	Share	Worst cell	Rows whose total the repair changed
first name by sex	5,552	10.4%	-9	0
first name by race	12,044	22.5%	-6	0

The published tables have no negatives; the Bureau also ships the un-repaired copies “for research purposes,” and comparing them shows what the repair did.

First name	People	Published	Research copy
RENITA	6,168	0 / 6,168	-1 / 6,169
SHERIE	4,529	0 / 4,529	-3 / 4,532
YAAKOV	4,229	4,229 / 0	4,230 / -1
KIRSTIE	3,655	0 / 3,655	-1 / 3,656
ALETHEA	3,568	0 / 3,568	-1 / 3,569
FATEMEH	3,568	0 / 3,568	-2 / 3,570
IZABELLE	3,515	0 / 3,515	-1 / 3,516
KYMBERLY	3,433	0 / 3,433	-1 / 3,434

Minus one man named RENITA. The repair moves the impossible count to zero and takes the difference out of the cell beside it, so **every row's total is preserved exactly** — 107,226 of 107,226 across both tables. The surnames chapter works through the same phenomenon in the last-name tables, where the counts are larger and the negatives rarer.

11 What this chapter cannot tell you

The census file itself is not about age, and the birth file brought in above only patches the hole. A first name is a fashion object, given at a moment in time, and the most useful thing you could know about DOROTHY is when it was given. The census tables are one snapshot of everyone alive in 2020 and cannot supply it. There is a construction detail on top of that limitation. To be published, a 2020 first name had to have occurred at least a hundred times **in both 2010 and 2020**. Names that arrived during the 2010s are absent by design.

And the birth file is not a birth file. It counts applications for a Social Security card, tabulated by the year of birth the applicant gave, and it counts the name written on the application rather than the edited name the census publishes. Numbers were first issued in 1936, so anyone born before then is in it only if they lived long enough to apply. Of the births the country registered in 1920, 77% reached the file, and by 1950 that was 96%. Before 1915 there is nothing to hold it against, because the country kept no national count of births. It also drops any name given to fewer than five babies in a year, and it knows nothing about anyone born outside the United States.

Nothing here is about geography. Names are not spread evenly across the country, and the entire premise of surname-based inference is that combining a name with a place does far more than either alone. Everything in this chapter is national.

There is no joint distribution. First names and last names are tabulated separately, always, and the Bureau says so explicitly. Every statement in this chapter about full names rests on multiplying two marginals, and the 12% correction computed above is a floor on the error in that step, not an estimate of it.

These are not national demographic statistics, and the Bureau prints that warning on both products. The racial composition of the people in this file is not the racial composition of the United States. The difference is the nine to ten percent of the census whose names could not be used.

Sex is not gender, and two boxes are not a spectrum. The 7,042,013 people the majority rule gets wrong are people whose recorded sex differs from their name's majority; nothing here counts people for whom the recorded sex is itself the wrong answer.

And the names are edited names. MICHAEL III is MICHAEL, PEÑA is PENA, middle names and suffixes are gone, and by the Bureau's own account as much as five percent of what is left is not a name at all. That editing is the right call for a frequency table. It is also the thing that makes this file a poor guide to what a raw name field looks like — which is what every chapter that matches records is actually working with.

12 Sources

Data

- U.S. Census Bureau. *Frequently Occurring First Names in the 2020 Census*, by sex and by race and Hispanic origin, together with the "WithNegatives" research copies of both. Four spreadsheets, no key or account required. https://www.census.gov/topics/population/genealogy/data/2020_names.html
- U.S. Census Bureau. *Frequently Occurring Last Names in the 2020 Census*, by race and Hispanic origin, from the same page. The surnames chapter uses this spreadsheet too and keeps its own copy; nothing here depends on that one.
- Comenetz, Joshua. *First Name Data From the 2020 Census*. 2020 Census Briefs C2020BR-13, April 2026, and *Last Name Data From the 2020 Census*, C2020BR-14. Every statement above about editing, generic entries, the question-mark substitution, the two-vintage threshold and the size of the infused noise comes from these two documents. <https://www2.census.gov/library/publications/decennial/2020/c2020br-13.pdf> and [c2020br-14.pdf](https://www2.census.gov/library/publications/decennial/2020/c2020br-14.pdf). Disclosure Review Board approval CBDRB-FY26-058.
- Social Security Administration. *Beyond the Top 1000 Names*, national data. One plain text file for every year of birth from 1880 to 2025, three columns: name, sex, count. Everything in the flow-and-stock section comes from here, and nothing else in the chapter does. <https://www.ssa.gov/oact/babynames/limits.html>
- National Center for Health Statistics. *Vital Statistics of the United States*, historical birth registrations. Four years only, 1920 through 1950, used as a yardstick for how much of each year the birth file caught.
- Nothing else. Every other number in this chapter comes from the two census tabulations and the Bureau's briefs on them. No voter file, survey or registry is read, and the collision table is arithmetic on the published counts rather than a count of matches in anybody's list.

The data itself

Every figure above is drawn from these tables. They are plain CSV, they open in a spreadsheet, and they are yours to take further.

`both_ways.csv`, `collision.csv`, `cover.csv`, `curve.csv`, `disagreement.csv`, `disagreement_top.csv`, `explore_first.csv`, `explore_last.csv`, `negatives.csv`, `negatives_top.csv`, `race_extremes.csv`, `race_info.csv`, `sex_bands.csv`, `sex_scatter.csv`, `ssa_alexandra.csv`, `ssa_births.csv`, `ssa_coverage.csv`, `ssa_explore.csv`, `ssa_newcomers.csv`, `ssa_risers.csv`, `ssa_shock.csv`, `ssa_totals.csv`, `ssa_vs_census.csv`, `top25.csv`, `unisex.csv`

Your turn

None of these is answered above, and every one can be answered from the tables you just opened.

- `explore_first.csv` gives every common first name a `pct_female` and a set of race shares. Sort by `pct_female` and find the names sitting near the middle. What happens to anyone who guesses sex from those names?
- `race_info.csv` reports how much uncertainty a name actually removes, for first names and last names. Compare the two. Which half of a person's name tells you more, and is either enough to act on?
- `negatives.csv` and `negatives_top.csv` show published counts that go below zero once the research copy is used. Find a name where the published figure and the research figure disagree. What was the Bureau protecting, and what did it cost you?
- `ssa_totals.csv` has `census_2020` beside `born_all` and `pct_still_counted`. Pick a name that was once common and is now rare. Work out whether it is dying out or just ageing.

Related work

- Word, David L., Charles D. Coleman, Robert Nunziata and Robert Kominski (2008). "Demographic Aspects of Surnames from Census 2000." U.S. Census Bureau technical documentation — the methodology these products descend from.
- Bertrand, Marianne and Sendhil Mullainathan (2004). "Are Emily and Greg More Employable Than Lakisha and Jamal? A Field Experiment on Labor Market Discrimination." *American Economic Review* 94(4): 991-1013 — the best-known demonstration that a first name carries a signal somebody acts on.
- Fryer, Roland G. and Steven D. Levitt (2004). "The Causes and Consequences of Distinctively Black Names." *Quarterly Journal of Economics* 119(3): 767-805.
- Elliott, Marc N. et al. (2009). "Using the Census Bureau's surname list to improve estimates of race/ethnicity and associated disparities." *Health Services and Outcomes Research Methodology* 9: 69-83 — the method the surnames and BISG chapters implement by hand.

How the figures were made

Every number above is computed at render time from the five spreadsheets, and none is typed into the prose. Two conventions are worth naming. The collision probabilities are computed over the **whole** population, residual line included, rather than over the published rows rescaled to themselves. Rescaling would drop the people with rare names, who are precisely the people least likely to collide, and would overstate every figure in the identification section. And every figure that carries a cut prints the coverage that cut buys, in the figure itself, so that a reader cannot mistake any of them for the whole table.

There are three: 2,000 names in the lookup, 25,000 bearers in the scatter, and 1,000 names in the birth curves.

Checks

Check	Value
first names published	53,615
last names published	156,621
people with a first name in the file	302,031,536
people with a last name in the file	298,898,231

Check	Value
race columns sum to the published total, every last name	156,621 of 156,621
sex columns sum to the published total, every first name	53,615 of 53,615
first names given two different totals by the two tabulations	49,089 of 53,615 (91.56%)
rows whose total the negative repair left unchanged	107,226 of 107,226
names that are both a first name and a last name	14,926
chance two people share both a first and a last name	1 in 1,212,277
birth years in the Social Security file	1880-2025 (146 years)
girls named ALEXA, launch year against the last year	6,053 in 2015, 243 in 2025
ALEXA between the two censuses: born, against counted alive	38,306 born, stock up 42.02%
names given to 1,000+ babies since 2023 that the census file has no row for	86 names, 100,993 babies

Each line is asserted before any of the tables above are written; a change in any source that moved one of these numbers would stop the document rather than change it quietly.

REBUILD THIS DATA YOURSELF, WITH AN AI ASSISTANT

I want to rebuild two public datasets about American names from their original sources, without any starting files. Please do the following and show your work.

THE FIRST SOURCE. U.S. Census Bureau, "Frequently Occurring First Names in the 2020 Census" and "Frequently Occurring Last Names in the 2020 Census", published April 2026. Spreadsheets, served plainly from <https://www2.census.gov/topics/genealogy/2020surnames/>

No account or key is needed. Fetch two of them:

Names2020_FirstNames_Sex.xlsx (about 3 MB) and

Names2020_LastNames_RaceHispanic.xlsx (about 11 MB). In each, two header rows sit above the column names, so skip those. Read every column as text first and convert yourself – some count columns arrive as text, and the proportion columns carry seventeen digits. Each file ends with one aggregate row called ALL OTHER NAMES: it is not a name, so leave it out when counting names, but it is people, so keep it when totaling how many people the file covers.

THE SECOND SOURCE. Social Security Administration, "Beyond the Top 1000 Names", national data: <https://www.ssa.gov/oact/babynames/names.zip> (about 8 MB) – a zip holding one text file per birth year, 1880 onward, each row a name, a sex, and a count of babies given that name.

HOW TO FETCH THE SECOND ONE. The Social Security server refuses plain scripted downloads. Send a browser user-agent header AND accept compressed responses – either alone still fails. If that argument cannot be won, download it by hand in a browser and start from the file.

WHAT TO BUILD.

1. From the census spreadsheets: how many distinct first names and how many distinct last names the Bureau published, and how many people each file covers.

2. From the Social Security files: the span of birth years, and the count of girls named ALEXA in 2015 against 2025.

CHECK YOUR WORK. The copies this book used were fetched in August 2026.

If your rebuild matches them, you should find:

- 53,615 first names and 156,621 last names published
- 302,031,536 people with a first name in the census file, and 298,898,231 with a last name
- birth years 1880 through 2025 in the Social Security file
- 6,053 girls named ALEXA in 2015, and 243 in 2025

The Social Security file grows by one year every May; a later top year means the file was extended, not that you made an error.